

Low-Light Image Enhancement with Cross-Dataset Generalization

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Abstract

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1. Introduction

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2. Experimental Protocol

2.1. Datasets

The pipeline developed in this work is based on four datasets, each with a specific role in the experimental protocol: (i) LOL-v2 [4], (ii) LOL-v1 [3], (iii) MILs [2], and (iv) ExDark [1]. All models are trained on a single source domain and are then evaluated on additional datasets without fine-tuning. This choice is intended to measure cross-dataset generalization, i.e., the ability of each model to enhance low-light images whose distribution differs from the training data.

The main training dataset is LOL-v2. This dataset contains both a real-captured subset, hereafter referred to as LOL-v2 Real, and a synthetic subset, referred to as LOL-v2 Synthetic. In the main protocol, only LOL-v2 Real is used, meaning that the low-light/normal-light pairs are acquired from real scenes rather than obtained through synthetic low-light degradation. LOL-v2 is a paired dataset: each low-light image is associated with a corresponding normally exposed reference image, which can be used as the target for supervised training and evaluation.

LOL-v1 is used only as a paired cross-dataset evaluation set. It represents a relatively close cross-dataset scenario, because it belongs to the same general benchmark family as LOL-v2.

MILs is used as a second paired cross-dataset evaluation set. In particular, the low-resolution DSLR split, MILs-DSLR, is adopted to keep the evaluation computationally manageable. Unlike LOL-v1, this dataset introduces a controlled illumination setting. Its input images are associated

with different illumination levels, and each input has a corresponding well-illuminated reference image. This makes it possible to evaluate the models not only with global results over the whole dataset, but also by studying how performance changes as the input illumination becomes more severe.

Finally, ExDark is used as an unpaired stress-test dataset. Here, unpaired means that the dataset provides low-light images without a corresponding normally exposed reference image. Therefore, ExDark is not used for reference-based comparison, but for no-reference evaluation and qualitative failure analysis on real-world dark scenes. Its category-based organization is preserved, meaning that images remain grouped according to the object category provided by the dataset, such as cars, people, animals, or indoor objects. This organization may help inspect whether some failure cases are related to the type of scene or object.

2.2. Split Construction

Split construction follows the role assigned to each dataset in the experimental protocol. For the main source domain, LOL-v2 Real, the official training split is used to derive the training and validation partitions. In particular, a fixed fraction of the official training samples is deterministically held out as validation data using a fixed random seed. The official test split is kept untouched for the final in-domain evaluation. In this context, “in-domain” means that training, validation, and testing images come from the same dataset and acquisition setting.

The synthetic subset of LOL-v2 is kept separate from the main protocol and is used only for an additional ablation study, aimed at evaluating whether synthetically degraded low-light examples improve or degrade generalization. When this subset is used, a validation partition is deterministically derived from the official training split using the same procedure adopted for LOL-v2 Real, ensuring consistency in the split construction process.

For the remaining datasets, no additional train or validation partitions are created. Instead, all available samples are assigned to a single evaluation partition. For datasets

Table 1. Dataset split sizes used in the pipeline. Counts refer to samples, i.e., low-light input images. For paired datasets, each input sample is associated with a corresponding target image.

Dataset	Train	Validation	Test
ExDark	–	–	7363
LOL-v1	–	–	500
LOL-v2 Real	620	69	100
LOL-v2 Synthetic	810	90	100
MILLs	–	–	499

that already provide predefined splits, such as LOL-v1 and MILLs, the original partitions are merged into a unified test set. ExDark is likewise converted into a single test split, while preserving the original category labels as metadata.

The resulting split sizes are summarized in Table 1.

3. End-to-End Pipeline

The experimental workflow is organized as a sequence of modular stages, in which each stage performs a specific task and produces outputs for the following stages, from the raw datasets to the final quantitative and qualitative evaluation. The planned pipeline consists of (i) manifest generation, (ii) offline preprocessing, (iii) model training, and (iv) model evaluation.

The only dataset-specific stage is manifest generation. Besides its natural role of organizing the data, this step is intentionally designed to convert datasets with different structures into a common format. As a result, the subsequent stages of the pipeline can operate on the same data representation without relying on the original organization of each dataset, making them effectively dataset-agnostic.

3.1. Manifest Generation

The purpose of the first stage of the pipeline is to scan the original dataset folders and build CSV manifest files from their structure. In each manifest, every row represents one image sample and stores the dataset name, the split to which the sample belongs, the path of the low-light input image, the path of the corresponding target image when available, and optional metadata such as the image category or the illumination level.

This step is necessary because the datasets used in this project are not organized in the same way. Some datasets provide paired low-light and normal-light images with the same filename, while others use different filename conventions. ExDark is instead unpaired and organized by object category. MILLs also includes an illumination level in the input filename, which is useful for analyzing how performance changes under different lighting conditions. Without this standardization step, the subsequent scripts in the pipeline would need to contain several dataset-specific rules.

For this reason, manifest generation is the only intentionally dataset-specific part of the pipeline. The script reads a YAML configuration file that describes where each dataset is located, which output file should be produced, and how the corresponding manifest should be created. In LOL-v1 and LOL-v2 Synthetic, low-light images and their corresponding reference images share the same filename, making the association straightforward. In LOL-v2 Real, the filenames differ only because low-light and reference images use different prefixes. ExDark follows a completely different structure: images are grouped by category and no reference image is provided, so only the image path and category information are recorded. Finally, in MILLs, the filename also contains information about the illumination level, which is extracted and stored together with the path of the corresponding reference image.

The output of this step is a CSV manifest file for each dataset, with a fixed column structure. This common format is what makes the later stages dataset-agnostic: once the manifests have been created, preprocessing, training, and evaluation only need to read the paths and metadata stored in the CSV files, without knowing how the original dataset was arranged on disk.

The manifest generation script also supports reproducibility-oriented choices. When a dataset does not provide an explicit validation set, a validation split can be sampled from the training split using a fixed random seed. Moreover, the generated rows are sorted according to the split and image path, making the row order deterministic across runs.

3.2. Offline Preprocessing

After manifest generation, all datasets are processed by a common offline preprocessing script. This stage is called “offline” because it contains deterministic preliminary operations, that are performed once rather than repeatedly during training. The processed images are stored on disk and are then used by both the training and evaluation stages.

The script reads the CSV manifests produced in the previous stage of the pipeline and creates a preprocessed version of each dataset on disk. For each dataset, it also writes a new CSV manifest with the same columns as the original one, but with paths updated to the preprocessed images.

Each image undergoes the following deterministic operations: (i) loading with EXIF orientation applied when available, (ii) conversion to RGB, (iii) resizing to 256×256 using Lanczos interpolation to estimate pixel values from neighboring pixels, and (iv) saving in PNG format. Applying the EXIF orientation avoids processing photographs with an incorrect rotation.

A fixed resizing strategy is used instead of cropping. In this project, preserving the whole scene during preprocessing was considered preferable to selecting a local region,

because cropping would permanently discard part of the image and could remove relevant low-light regions, light sources, objects, or failure cases needed for cross-dataset analysis. Although resizing may slightly distort the aspect ratio of some images, it keeps the full scene and maintains a fixed deterministic input size for all datasets. Random crops and other data augmentation operations are therefore left to the training stage, where they can be applied only to the training split.

Before processing a manifest, the script performs a lightweight consistency check. It verifies that no input path is duplicated within the same split and that the same input path does not appear in different splits. These checks are intentionally limited, since the dataset-specific pairing rules have already been handled during manifest generation.

The script also computes simple dataset-level statistics before applying the final resize. For each image of size $H \times W$, a luminance map $Y \in [0, 1]^{H \times W}$ is computed from the RGB channels using the ITU-R BT.709 luma coefficients:

$$Y_{i,j} = 0.2126R_{i,j} + 0.7152G_{i,j} + 0.0722B_{i,j}, \quad (1)$$

where $R_{i,j}$, $G_{i,j}$, and $B_{i,j}$ are the RGB values of pixel (i, j) , scaled to $[0, 1]$. Based on this luminance map, the dark pixel ratio is defined as

$$\text{DPR} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W 1[Y_{i,j} < \tau_d], \quad \tau_d = 0.10 \quad (2)$$

where H and W are the image height and width respectively, τ_d is the dark-pixel threshold, and $1[\cdot]$ is the indicator function, that is equal to one when the condition is true and zero otherwise. For paired datasets, the same luminance statistics are also computed on the target images. These statistics are summarized in Table 2 and provide a compact description of the illumination distribution of each dataset. The table shows that LOL-v1 and LOL-v2 Real have very similar input luminance distributions, supporting the use of LOL-v1 as a close cross-dataset benchmark for the main training domain. LOL-v2 Synthetic and MILs are instead brighter on average, indicating a larger distribution shift in terms of illumination. ExDark lies between these cases in terms of input luminance and remains useful as an unpaired real-world stress test.

4. Conclusion

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References

- [1] Yuen Peng Loh and Chee Seng Chan. Getting to know low-light images with the exclusively dark dataset. *Comput. Vis. Image Underst.*, 178:30–42, 2019. [1](#)
- [2] Maria Pilligua, David Serrano-Lozano, Pai Peng, Ramon Baldrich, Michael S. Brown, and Javier Vazquez-Corral. Evaluating low-light image enhancement across multiple intensity levels. *CoRR*, abs/2511.15496, 2025. [1](#)
- [3] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. In *BMVC*, page 155. BMVA Press, 2018. [1](#)
- [4] Wenhan Yang, Wenjing Wang, Haofeng Huang, Shiqi Wang, and Jiaying Liu. Sparse gradient regularized deep retinex network for robust low-light image enhancement. *IEEE Trans. Image Process.*, 30:2072–2086, 2021. [1](#)

Table 2. Dataset statistics computed from the preprocessed datasets before the final resize. Luminance values are reported as mean $\pm$ standard deviation across samples. Dark pixels denote the average fraction of input pixels with luminance below 0.10.

Dataset	Samples	Median original size	Input luminance	Dark pixels	Target luminance
ExDark	7363	640×500	0.143 ± 0.086	59.6%	–
LOL-v1	500	600×400	0.061 ± 0.041	82.7%	0.458 ± 0.102
LOL-v2 Real	789	600×400	0.062 ± 0.042	80.9%	0.436 ± 0.105
LOL-v2 Synthetic	1000	384×384	0.240 ± 0.138	46.7%	0.513 ± 0.141
MILLs	499	600×400	0.244 ± 0.110	16.4%	0.369 ± 0.095